V.L Memorial Public School

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CHAPTER-3-QUESTIONS-CLASS8TH-SCIENCE

Class 08 - Science (Curiosity)

Section A

1. Which household device does NOT work on the heating effect of electric current? **[1]**

   a) Electric stove
   b) Electric room heater
   c) Electric fan
   d) Hair dryer

2. Why does a nichrome wire get warm when electric current passes through it? **[1]**

   a) It glows like a bulb filament
   b) Because resistance converts some electrical energy into heat energy
   c) It is too short to stay cool
   d) Due to being made from copper

3. What happens to the compass needle when current flows through a nearby wire? **[1]**

   a) It deflects
   b) It points to South
   c) It points to North
   d) It stays still

4. What happens to an electromagnet when the electric current flowing through it is switched off? **[1]**

   a) Its magnetic field disappears
   b) Its magnetic field reverses
   c) Its magnetic field increases
   d) Its magnetic field remains

5. Which pair of metals did Luigi Galvani use while observing the movement of a frog's leg? **[1]**

   a) Zinc and copper
   b) Iron and silver
   c) Copper and iron
   d) Aluminium and copper

6. Why is an iron core used in an electromagnet? **[1]**

   a) To strengthen magnetism
   b) To reduce current
   c) To reverse polarity
   d) To stop deflection

7. Which of the following appliances is based on the magnetic effect of electric current **[1]**

   a) Electric iron
   b) Water heater
   c) Electric stove
   d) Electric bell

8. Why is a dry cell considered a single-use cell? **[1]**

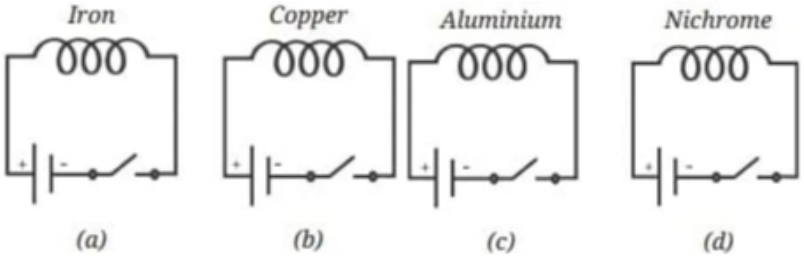
   a) It requires sunlight
   b) It contains liquid
   c) It cannot be recharged
   d) It uses two electrodes

9. Why should wires, plugs, and sockets in household switchboards have appropriate **[1]**

current ratings?

a) To increase socket weight
b) To prevent unnecessary heating
c) To reduce appliance size
d) To increase wire length

10. We have four coils, of similar shape and size, made up from iron, copper, aluminium, and nichrome as shown in Fig. When current is passed through the coils, compass needles placed near the coils will show deflection. **[1]**



When current is passed through the coils, compass needles placed near the coils will show deflection.

a) In all four circuits
b) Only in circuits (a) and (b)
c) Only in circuits (a), (b), and (c)
d) Only in circuit (a)

11. Voltaic cell uses _____ electrolyte that is why it is not considered as convenient as a dry cell. **[1]**

a) coloured
b) paste
c) solid
d) liquid

12. What type of compass needle movement is observed when electric current is stopped? **[1]**

a) It starts vibrating
b) It returns to its original direction
c) It spins continuously
d) It keeps deflecting

13. Why are dry cells preferable for everyday use compared to Voltaic cells? **[1]**

a) Dry cells are costly.
b) Dry cells do not use liquid electrolytes.
c) Dry cells have a larger size.
d) Dry cells are heavier.

14. What is the difference between Voltaic cells and Dry cells? **[1]**

a) Both use liquid electrolytes
b) They are identical in structure and function
c) Voltaic cells use a solid electrolyte; Dry cells use a liquid one
d) Voltaic cells use a liquid electrolyte; Dry cells use a paste

15. The heating effect of the electric current does NOT change with the change in **[1]**

a) Time duration for which the current is passed
b) Strength of the electric current
c) The nature, length, and thickness of the conductor
d) Direction of the flow of electric current

16. When creating a homemade battery, how can an individual extend the battery life? **[1]**

a) Increase the thickness of metal electrodes.
b) Use stronger acids in higher concentrations.
c) Select metals that have high oxidation rates.
d) Keep acid or salt solution concentrations balanced.

17. What is a common use of rechargeable batteries? **[1]**

a) Used in non-electronic products
b) Used multiple times after charging
c) Used only in medical equipment
d) Used once and then disposed of

18. What is the liquid used in a Voltaic cell called? **[1]**

a) Insulator
b) Semiconductor
c) Electrolyte
d) Conductor

19. In a dry cell, the positive terminal is the metallic tip on top of the carbon rod. Which then is the negative terminal of the cell? **[1]**

a) Bottom of the zinc container
b) Carbon rod just below the metallic tip
c) The paste of chemicals inside the cell
d) Bottom of the carbon rod

20. Who discovered the link between electricity and magnetism? **[1]**

a) Albert Einstein
b) Hans Christian Oersted
c) Isaac Newton
d) Thomas Edison

21. **Assertion (A):** Electromagnets do not have poles like the North pole and South pole of a bar magnet. **[1]**
Reason (R): When we interchange the terminals of the battery in a circuit, the poles of the electromagnet also change positions.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

22. **Assertion (A):** When electric current flows through a conductor (like a wire), it produces a magnetic field around it. This phenomenon is known as the magnetic effect of electric current. **[1]**
Reason (R): The magnetic field around a current-carrying conductor can be shown by the deflection of a compass brought near the conductor.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

23. **Assertion (A):** Dry cells are more convenient than Voltaic cells for portable use. **[1]**
Reason (R): Dry cells use a thick moist paste, avoiding spillage issues associated with liquid electrolytes.

a) Both A and R are true and R is the
b) Both A and R are true but R is not

correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

24. **Assertion (A):** Electric vehicles heavily rely on lithium-ion batteries. **[1]**
Reason (R): Lithium and cobalt, key components of these batteries, are sourced from limited global locations.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

25. **Assertion (A):** Nichrome wire heats up more when an electric current is passed through it compared to copper wire of the same size and length. **[1]**
Reason (R): Nichrome wire offers higher resistance than copper wire, causing more electrical energy to be converted to heat.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

26. **Assertion (A):** The filament in an incandescent lamp glows due to the heating effect of electric current. **[1]**
Reason (R): The filament is made of copper wire which offers low resistance, converting electrical energy to heat.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

27. **Assertion (A):** Rechargeable batteries remain effective indefinitely. **[1]**
Reason (R): Over time, rechargeable batteries wear out after multiple charge and discharge cycles.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

28. What is the observed effect when a nichrome wire is connected to a current source? **[1]**
29. Why should wires, plugs, and sockets in household switchboards have appropriate current ratings? **[1]**
30. Identify the scientist who discovered the link between electricity and magnetism in 1820. **[1]**
31. What phenomenon is described when an electric current produces a magnetic field around a conductor? **[1]**
32. What is meant by the heating effect of electric current? **[1]**
33. Describe how the strength of an electromagnet can be increased. **[1]**
34. What is an electromagnet? **[1]**
35. Explain why solid-state batteries are a significant development in battery technology. **[1]**

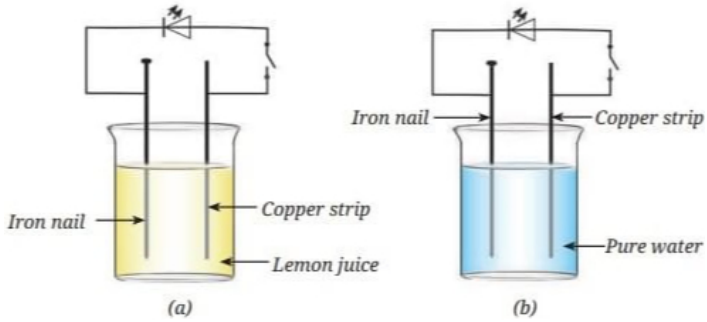
Section B

36. Match the items in Column A with those in Column B. **[2]**

Column A	Column B
(i) Voltaic cell	(a) Best suited for electric heater
(ii) Electric iron	(b) Works on magnetic effect of electric current
(iii) Nichrome wire	(c) Works on heating effect of electric current
(iv) Electromagnet	(d) Generates electricity by chemical reactions

37. Describe an application of electromagnets and how they benefit from the magnetic effect of electric current. **[2]**

38. In Fig., in which case the LED will glow when the switch is closed? **[2]**



39. What creative solution is suggested for preventing energy loss and overheating in electrical circuits? **[2]**

40. In the context of preventing overheating, what considerations should be made when choosing household wiring and sockets? **[2]**

41. Describe the principle that causes household appliances to generate heat when powered by electric current. **[2]**

Section C

42. Explain the reasons nichrome wire is used as a heating element in various electrical appliances based on its physical and electrical characteristics. **[3]**

43. Evaluate the implications of electromagnets compared to permanent magnets in industrial applications, focusing on their advantages. **[3]**

44. What is the main difference between Voltaic cells and dry cells? **[3]**

45. How does the use of two cells affect the heat generated in a wire, and what factors determine the amount of heat produced? **[3]**

46. Discuss the importance of selecting appropriate materials for household wiring to minimize unnecessary heating and energy loss. **[3]**

Section D

47. **Read the following text carefully and answer the questions that follow:** **[4]**

When an electric current passes through a straight conductor, it produces a magnetic field around it. The direction of this magnetic field can be found using the Right-Hand Thumb Rule. According to this rule, if you hold the current-carrying conductor in your right hand with the thumb pointing in the direction of the current, then the curl of the fingers gives the direction of the magnetic field around the conductor.

Questions:

a. How do you find the direction of the magnetic field around a current-carrying straight conductor? **(1)**

b. What does the Right-Hand Thumb Rule state? **(1)**

c. How can you demonstrate the direction of the magnetic field around a conductor using the Right-Hand Thumb Rule? **(2)**

OR

Why is the Right-Hand Thumb Rule important in understanding electromagnetism? **(2)**

48. **Read the following text carefully and answer the questions that follow:** **[4]**

When electric current flows through a conductor, it faces resistance, causing electrical energy to change into heat. This is called the heating effect of electric current. Nichrome wire, which has high resistance, heats more than copper wire of the same size. The amount of heat generated depends on the current, material, length, thickness of the wire, and duration of flow. This principle is used in devices like heaters, irons, and stoves.

Questions:

a. Which factor does not affect the heating effect of electric current? Explain briefly. **(1)**

b. What is the heating effect of electric current? **(1)**

c. Why does a nichrome wire get heated more than a copper wire of the same size? **(2)**

OR

Explain one advantage and one disadvantage of the heating effect of electric current. **(2)**

49. **Read the following text carefully and answer the questions that follow:** **[4]**

An electromagnet is formed when a coil of wire is wound around a material like soft iron, and current is passed through it. The coil produces a magnetic field, and the soft iron core becomes strongly magnetized. The magnetism remains as long as the current flows but disappears once the current is switched off. Electromagnets are widely used in devices such as electric bells, cranes for lifting heavy scrap iron, and electric motors.

Questions:

a. What happens to the magnetism of an electromagnet when the electric current is switched off? **(1)**

b. Why does soft iron make a good core for an electromagnet? **(1)**

c. How does an electromagnet in a crane help in lifting heavy scrap iron? **(2)**

OR

Differentiate between a permanent magnet and an electromagnet. **(2)**

Section E

50. Answer the following: **[5]**

i. Differentiate between the structure of a Voltaic cell and a dry cell.

ii. Identify the negative terminal of a dry cell.

iii. Explain why dry cells are more convenient for everyday use.

51. Identify and explain the historical impact of the invention of the Voltaic cell on scientific innovation and industrial technology. **[5]**

52. How can electromagnets be made stronger while retaining their temporary nature? **[5]**

53. Analyze why safety devices are critical in household circuits where electric current heating is in effect. **[5]**

54. Answer the following: **[5]**

i. Evaluate the advantages of rechargeable batteries.

ii. State whether rechargeable batteries last forever.

iii. Explain why solid-state batteries are considered a possible future improvement.

55. Answer the following: **[5]**

i. Explain how Earth's magnetic field is produced.

ii. State one way in which living organisms use Earth's magnetic field.

iii. Explain how Earth's magnetic field helps protect life on Earth.

Section F

56. **State whether the given statement is True or False:** **[10]**

(a) The heat generated in the nichrome wire depends on the material, thickness, and length of the wire. **[1]**

(b) Using properly rated wires can eliminate all risks associated with the heating effect of electric current. **[1]**

(c) Pure water can be used as an electrolyte in a Voltaic cell. **[1]**

(d) The heating effect of electric current depends on the direction of the flow of electric current. **[1]**

(e) Manufacturing steel often uses the chemical effect of electricity to melt and recycle steel. **[1]**

(f) If you increase the number of cells in a circuit, the nichrome wire will heat up more. **[1]**

(g) Dry cells use solid electrolytes; therefore are convenient to carry anywhere. **[1]**

(h) Dry cells use a paste-like electrolyte instead of a liquid. **[1]**

(i) Lithium-ion batteries use lithium and cobalt, which are plentiful worldwide. **[1]**

(j) A lifting electromagnet cannot release the objects once it has picked them up. **[1]**

57. **Fill in the blanks:** **[10]**

(a) The heat generated is greater when using a battery of _____ cells instead of one. **[1]**

(b) When we increase the current flowing through the coil (use battery of two or more cells in place of one cell), the magnet becomes _____. **[1]**

(c) The scientist who discovered the magnetic effect of electric current was _____. **[1]**

(d) The polarity of an electromagnet can be reversed by changing the _____ of the current. **[1]**

(e) Solid-state batteries replace liquid electrolytes with _____. **[1]**

(f) A current carrying coil that behaves as a magnet is called an _____. **[1]**

(g) The solution used in a Voltaic cell is called _____. **[1]**

(h) An incandescent lamp glows because its _____ is heated by an electric current. **[1]**

(i) When electric current flows through a conductor like a wire, it produces a **[1]**

_____ around it.

(j) If the direction of the flow of _____ is reversed, the poles of electromagnet interchange their positions. **[1]**